



Pop filters for mics

If you want to prevent the Darth Vaderesque breathing noises on your mics, use these pop filters! <https://digit.in/aug20-33>



Reading goes virtual

Authors are moving to web novels and audiobooks, at a time when bookstores are closed. <https://digit.in/aug20-34>

instruments, “(The experience was) excellent. Without these instruments, we wouldn’t have been able to what we do. We also use the KEMAR dummy-head, which, to our knowledge, is one of only two in all of India”.

The RR1 Conquest, just as all electrostatic headphones, requires a special ‘Bias’ voltage amplifier to drive it. While Kaldas Research doesn’t make these amplifiers themselves, there are plenty of options available in the market. According to Chandan, “Some of the world’s best E-stat (electrostatic) amps are DIY, such as the KGSSHV”.

WHY ELECTROSTATIC?

For Kaldas Research, the reason to choose electrostatic transmission was a personal one. Chandan stated, “We were not interested in electrodynamic as we wanted to keep everything in-house and the R&D cost for an in-house dynamic driver is astronomical since a lot of specialised equipment is re-



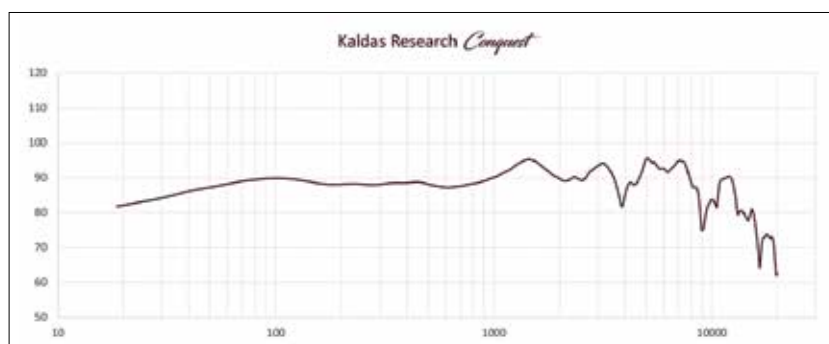
The driver of the RR1 Conquest

quired”. According to him, Planar technology is ubiquitous, so they weren’t interested in that approach as well.

Interestingly, electrostatic wasn’t Kaldas Research’s first choice for the RR1 Conquest. “We had something else in development, another form of transmission that is a few years away from being commercially feasible. There are still a few idiosyncrasies to overcome there”, said Chandan. Due to this, the company finally opted for electrostatic.

THE UNIQUE BUILD AND DESIGN

The RR1’s chassis consists of several materials. The headband assembly is



RR1 Conquest uncompensated Frequency Response graph

made out of CNC-milled Aluminium, the headband harp is spring steel, and the hardening process is done by Kaldas Research. The initial batches of the RR1 Conquest had 3D printed earcups. However, Chandan divulged that this proved to be a very demanding task that required a lot of manual processing to ensure a smooth surface. So, they shifted to full injection thermoplastic housings that are manually processed to get a matte-black finish.

The headphones also have an eye-catching design nuance – exposed wires! As mentioned before, the company was experimenting with a completely new driver type before they chose electrostatic. The experimental driver caused the initial prototypes to have an angled shaped, so instead of completely scrapping their design choice (which they loved), the engineers incorporated an electrostatic driver within the confines of the prototype’s design. The result? This edgy-looking design with exposed wiring!

Chandan further elaborated, “The exposed wiring is done in a way that is completely safe. Everything is insulated and there is no way to harm the device or yourself”.

THE SOUND PROFILE

Electrostatic headphones have minimal excursion (displacement of drivers) due to the lack of moving parts. Therefore, headphones employing electrostatic transmission are known to have understated bass response and slightly high treble. However, Chandan wanted to attempt to play around with the low-frequency characteristics with the

RR1 Conquest, without messing with the mid and high frequencies. While electrostatic headphones do have limited excursion, the correct mechanical tuning can change that.

The RR1 Conquest, reportedly, manages to extract quite a bit of bass response in comparison to traditional electrostatic headphones. The sound profile, as per Chandan, is slightly darker than other electrostatic headphones and it sounds quite different from what you’d be used to with an electrostatic headphone. Chandan doesn’t think of the extended bass response as an improvement since he believes that audio is highly subjective. Rightly so, since the process of bringing objectivity into sound quality has historically been tough, and even subjective testing achieved from systems such as G.R.A.S. cannot account for individualistic tastes.

LOOKING TO THE FUTURE

Kaldas Research’s next product line is called ‘Cetacean’, which isn’t another audio product but a line of tools and instruments made from Titanium that are mainly used in the precision tool industry. When asked about future plans in the audio realm, Chandan said, “I won’t be able to comment on that as we prefer for the finished work to talk for itself and currently, things are under development”. We, for one, are excited to see what Kaldas Research does next in the audio realm. The RR1 Conquest, however, is a major step towards India cementing itself as a player in the high-end audio manufacturing industry. 📌